

Project Report: TaskBridge Assessment

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Project Name: TaskBridge

Objective: To build a simple and clean full-stack web application to manage daily tasks and directly import tasks from Google Sheets.

1. Technologies Used (MERN Stack)

- **Frontend:** React.js, Tailwind CSS (for UI design), Axios (for API calls).
- **Backend:** Node.js, Express.js.
- **Database:** MongoDB.
- **Extra Package:** `csv-parser` (to read data from Google Sheets).

2. Main Features

1. **Google Sheet Import:** Users can paste any public Google Sheet link and import tasks directly into the database.
2. **Duplicate Prevention:** If a task with the exact same title and due date has already been imported from the sheet, the backend prevents it from being saved again.
3. **Task Management (CRUD):**
 - **Create:** Add a new task manually (Title, Description, Due Date).
 - **Read:** View the complete list of tasks, filter them (All, Pending, Completed), and sort them.
 - **Update:** Edit a task or mark it as "Completed".
 - **Delete:** Remove any task from the list.

3. How the Project Works (Data Flow)

A. Normal Task Flow:

- The user enters task details in the UI form (`TaskForm`).
- The data is sent to the backend (`POST /tasks`) and saved in MongoDB.
- The task list updates automatically on the UI after saving.

B. Google Sheet Import Flow:

1. The user pastes the spreadsheet URL in the `ImportSheet` section.
2. The backend extracts the unique Sheet ID from the URL and fetches the data in CSV format from Google.
3. The backend reads the rows (Title, Description, Due Date) using `csv-parser`.
4. It checks the database to ensure there are no duplicate tasks being imported.
5. Valid tasks are saved in MongoDB, and a success message is displayed on the UI.

4. Simple Folder Structure

Plaintext

```
TaskBridge/
├── backend/
│   ├── controllers/
│   │   └── taskController.js (Contains all the main logic)
│   ├── models/
│   │   └── taskModel.js (Database schema structure)
│   ├── routes/
│   │   └── taskRoutes.js (API endpoints)
│   ├── server.js (Main file to start the backend)
│   └── package.json
└── frontend/
    ├── src/
    │   ├── components/
    │   │   ├── ImportSheet.jsx
    │   │   ├── TaskForm.jsx
    │   │   └── TaskList.jsx
    │   ├── App.jsx
    │   └── main.jsx
    └── package.json
```

5. Setup Instructions (How to Run)

Backend:

1. Navigate to the backend folder.
2. Run the `npm install` command.
3. Create a `.env` file and add your database string: `MONGODB_URI=your_database_link`
4. Run `node server.js` (The server will start on Port 5000).

Frontend:

1. Navigate to the frontend folder.
2. Run the `npm install` command.
3. Run `npm run dev` to start the frontend application.

6. Project Links

- **Live Project Link:** <https://taskbridge-task-management-system.onrender.com/>
- **GitHub Code Link:** <https://github.com/jitesh-saini-dev/taskbridge-task-management-system>